Construction of new ovoid coordinate systems to describe ovoid spacetimes

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Cartesian ovals are a generalization of conic curves that have been proved to be useful to construct optical systems free of geometrical and chromatic aberrations. Inspired in that, in this work, new orthogonal curvilinear coordinates based on Cartesian ovals are constructed. To do so, the orthogonal curves to certain families of ovals are found. For confocal ovals, the orthogonal curves turn out to be a new class of curves, which are referred to as synecdolas, and for triconfocal ovals they are another family of ovals. Then, from the equations associated to these curves, the new coordinates are defined. This process results in two new 2D orthogonal curvilinear coordinates with ovals as one of the coordinate curves. By "rotating" or "translating" the coordinates, cylindrical- or rotational-like 3D coordinates are obtained. Four coordinate systems are finally constructed. As a special case, some Pascal limaçon coordinates are found.

Two special families of ovals are consider to do so: confocal ovals and triconfocal ovals. The orthogonal curves to confocal ovals, which are referred to as synecdolas, are constructed. Then, new coordinates are inferred from the free parameters associated with each of these intersecting curves. Meanwhile, from the triconfocal ovals only, two new coordinates, that define orthogonal sets of curves, are inferred. Afterwards,

the resulting 2D coordinates are generalized to 3D ones by constructing cylindrical- or rotational-like coordinates.

Some Pascal limaçon coordinates are found to be a special case of triconfocal oval coordinates. Finally, a stationary ovoid metric ansatz is proposed from the Euclidean line element in confocal ovoid coordinates based on analogue procedure to the one used by Krasiński to construct ellipsoidal spacetimes. Static anisotropic fluid solutions to the Einstein field equations are found using this ansatz. This work sets the bases of a framework that helps to describe objects with an intermediate shape between a sphere and an ellipsoid.

I. INTRODUCTION

Cartesian ovals are a generalization of conic curves which have been prove to be useful to the optics. Originally, they are found as the solution to the problem of rigorous stigmatism, in which the light rays emitted from a punctual source are needed to be refracted to a single point throw an optical surface, which turns out to be a Cartesian ovoid. Over a plane containing the focal axis, one obtains a Cartesian oval as the interface between the two media separated by the optical surface. In recent years, they have been recover by Silva-Lora and Torres [1–4]. This authors use Cartesian ovoids to construct singlet lenses with aplanatism and lenses with many Cartesian surfaces with achromatic stigmatism, i.e., stigmatic systems free of chromatic aberration. This results allow to construct optical systems with better performance in forming high-resolution and colour images.

Intrigued by that good results, this work proposes that Cartesian ovals can also be important in the study of curvilinear coordinates, in particular in the study of exact solutions to partial differential equations such as the Laplace equation and to the Einstein field equations of general relativity. It is known that the searching of new orthogonal coordinates lead to new solutions to partial differential equations by the use of the method of separation of variables [5–7]. In general relativity, new system of coordinates could potentially lead to

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discover or rediscover exact solutions to the Einstein's field equations [8].

Cartesian ovals are also known in the literature for having a very rich set of properties. After their introduction in geometry by Descartes [9], it is discover by Chasles that any oval can be described in terms of three foci [10]. By the same time, Quelet [11] finds that ovals can be obtained by projecting the intersection curve of two cones with parallel axes over certain plane and that ovals are the involute of certain caustic by refraction of a circle. It is also known that the inversion of an oval on a circle is also an oval [12]. Limaçons of Pascal are also a special case of ovals [12]. Ovals are also related to the Weirtrass elliptic function $\wp(z)$ [13]. As a bonus, Cartesian ovals appear in an early article by James Clerk Maxwell and as a part of an proposed exercise in the Principia mathematica of Newton.

This article is divided as follows:

II. DESCARTES OVALS

A Descartes oval is defined as the locus of points in the plane for which the sum, $cr + c'r'$, of distances to two fixed points, r and r', multiplied by real constants, c and c', is constant [9, 12, 14, 15]. For the curve to exits, c and c' can not be simultaneously zero. Without loss of generality it can be considered $c \neq 0$, so that the equation of Descartes ovals can be written as

$$r + ar' = u\,, \tag{1}$$

with a and u real constants. A well known result about ovals is that there exits a set of three distinguishable bipolar equations that give rise to the same curve [10, 12, 14]. In this case, the additional two equations to equation 1 are

$$\text{sign}(\tau(1-a^2))ar'' + \frac{u}{\tau}r = (\tau + \xi)\,, \tag{2}$$

$$\text{sign}(\tau(1-a^2))r'' - \frac{u}{\tau}r' = a\xi\,, \tag{3}$$

where the two fixed points are considered to be in the x-axis of the xy-plane, so that if the fixed point related to r is at the origin then the other one is at $(0, \tau)$, with τ a real constant, the real constant ξ is equals to $\frac{u^2-\tau^2}{(1-a^2)\tau}$, r'' is the distance to a third fixed point at $(0, \tau + \xi)$, and $\text{sign}(x) := x/|x|$ is the sign function. This third fixed point is considered as a *third focus* [14, 15], so that any oval is related to three foci. Equations 2 and 3 give rise to the same curve as equation 1 provided that ξ is not infinite and that $\tau + \xi$ is not zero, in which cases one, or both, of the equations 2 and 3 become undefined.

A less known result, not typically mentioned, is that the curve given by the equation 1 is not empty when

$$u \geq \min(1, a)|\tau| \quad \text{if} \quad a > -1\,, \tag{4}$$

$$u \leq |\tau| \quad \text{if} \quad a < -1\,, \tag{5}$$

$$-|\tau| \leq u \leq |\tau| \quad \text{if} \quad a = -1\,, \tag{6}$$

where $\min(1, a)$ is the minimum value between 1 and a. These are the existence conditions of ovals. In the case $a > -1$, these results coincides with [15]. In the appendix X it is shown a proof of these conditions. Although the existence conditions are only explicitly mentioned in [15], as far as the authors know, they are in fact implicit in the work of Descartes [9]. As ovals are classified in genres, based on the procedure to draw them, a certain oval may exits

only if it is possible to construct them with ruler and compass, i.e., if the required distances and radii are positive, if the intersections between lines exists, if the lines can be drawn, etc.

III. ORTHOGONAL SYSTEMS OF COORDINATES BASED ON OVALS

Consider the rectangular Cartesian coordinates x, y in the xy-plane. Any two coordinates q^1, q^2, related to the rectangular ones by the equations $x = x(q^1, q^2)$ and $y = y(q^1, q^2)$, are said to be orthogonal if their metric coefficients,

$$g_{ij} = \frac{\partial \mathbf{r}}{\partial q^i} \cdot \frac{\partial \mathbf{r}}{\partial q^j}\,, \tag{7}$$

form a diagonal matrix, where $\mathbf{r} = x(q^1, q^2)\hat{\boldsymbol{\imath}} + y(q^1, q^2)\hat{\boldsymbol{\jmath}}$ is the position vector of the point (x, y) in the xy-plane and $\hat{\boldsymbol{\imath}}, \hat{\boldsymbol{\jmath}}$ are the unitary Cartesian vectors. It is not difficult to see that the orthogonality condition for q^1, q^2 is

$$x_{,1}x_{,2} + y_{,1}y_{,2} = 0\,, \tag{8}$$

where the subscripts indicate partial differentiation, $x_{,i} := \partial x/\partial q^i$ for each $i = 1, 2$.

This condition is equivalent to say that the q-constant curves, called the coordinate curves, intersect each other at right angles, as it is shown in the appendix X. Hence, to construct a 2D system of coordinates, it is sufficient to have a pair of uniparametric families of curves $\mathscr{F}_1 = \{\gamma_r\}_{r\in\mathcal{I}}$ and $\mathscr{F}_2 = \{\sigma_s\}_{s\in\mathcal{J}}$ so that if $\gamma \in \mathscr{F}_1$ and $\sigma \in \mathscr{F}_2$ then γ and σ meet at right angles. The parameters of these two families r, s may be taken as the q^i-coordinates. Naturally, to any point in the xy-plane must pass one and only one member of each family.

If, in addition, these families $\mathscr{F}_1, \mathscr{F}_2$ can be seen as the level curves of two functions f, g, respectively, then $r = f(x, y)$ and $s = g(x, y)$ are the relations between Cartesian and rs-coordinates. Their inversion, $x = x(r, s), y = y(r, s)$, would allow to compute the metric coefficients in terms of r and s. To construct *oval* coordinates, one may consider that one, or both, of the coordinate curves are ovals. In the next sections, two special families of ovals are considered: confocal ovals and triconfocal ovals. With them, various orthogonal coordinates will be built.

IV. CONFOCAL OVALS AND SYNECDOLAS

Consider the set of ovals defined by the level curves of the function

$$f(x,y) = \sqrt{x^2+y^2} + a\sqrt{(x-\tau)^2+y^2}\,, \tag{9}$$

with a and τ real constants. As this ovals share the same focal distance $|\tau|$, this class of ovals can be called as *confocal ovals*. Consider some curve described by the equation $y = y(x)$ such that it intersects orthogonally to any member of the family of level curves of f. It is not difficult to show that the condition of orthogonality is that

$$\frac{dy}{dx} = \frac{f_x}{f_y}\,, \tag{10}$$

where the subscripts in f indicate partial differentiation. In its differential form, this equation takes the form $f_y dx - f_x dy = 0$, which is an inexact differential equation. However it can be reduced to an exact equation by the integration factor $w = 1/y$, so that $(f_y/y)dx - (f_x/y)dy = 0$ is exact. Hence, it can be shown that the implicit solution to the equation 10 is

$$g(x,y) = \operatorname{arcsinh}\left(\frac{x}{y}\right) + a\operatorname{arcsinh}\left(\frac{x-\tau}{y}\right) = v\,, \tag{11}$$

with v a real constant. The set of level curves of g will be called *synecdolas*. Each of this curve crosses each confocal oval at a right angle. For $a = 1$ they become sections of hyperbolas, and for $a = -1$ they become upper or lower halves of ellipses. The term *synecdola* comes from the figure of speech "synecdoque". The intention to coin this term is inspired in the closeness of names of conics to figures of speech, such as in: circle and circumlocution, ellipse and ellipsis, hyperbola and hyperbole, and parabola and parable.

Synecdolas and confocal conform a pair of families which curves mutually intersect each other at 90 degrees. The inverse relations $x = x(u,v)$ and $y = y(u,v)$, obtained by solving the equations $f(x,y) = u$ and $g(x,y) = v$ for x and y, provide the relation between Cartesian coordinates and the new coordinates u and v. To find such inverse relations start by substituting the equations

$$x = \frac{\tau\sinh t}{\sinh t - \sinh\left(\frac{v-\tau}{a}\right)}\,, \tag{12}$$

$$y = \frac{\tau}{\sinh t - \sinh\left(\frac{v-\tau}{a}\right)}\,. \tag{13}$$

Then, the equation $g(x,y) = v$ becomes $v = v$, so that it is satisfied automatically by the substitution. The remain equation becomes

$$\frac{|\tau|\left(\cosh t + a\cosh\left(\frac{v-t}{a}\right)\right)}{\left|\sinh t - \sinh\left(\frac{v-t}{a}\right)\right|} = u\,. \tag{14}$$

The expression in absolute values is equal to $|y|$, with y taken from equation 13. Hence, by looking for inverse relations with $y = y(u,v) > 0$, equation 14 becomes

$$(a\tau - u)s^{\frac{2}{a}} + e^{\frac{v}{a}}(\tau - u)s^{\frac{1}{a}+1} + e^{\frac{v}{a}}(\tau + u)s^{\frac{1}{a}-1} + e^{\frac{2v}{a}}(a\tau + u) = 0\,, \tag{15}$$

where $s = e^t$. For $a \in \mathbb{Q}$ this equation can be rearranged into an algebraic equation of certain degree (see Appendix X). For $a = 1/3$, this equation turns out to be a cubic equation in s^2 whose positive real solution for s is

$$s = \sqrt{\alpha_+^{\frac{1}{3}} + \alpha_-^{\frac{1}{3}} - \frac{(\tau - u)e^{3v}}{\tau - 3u}}\,, \tag{16}$$

where

$$\begin{aligned}\alpha_\pm = \pm\,&\frac{e^{\frac{9v}{2}}}{(\tau - 3u)^2}\sqrt{\left[(\tau+3u)(\tau-u)^3e^{3v} - (\tau-3u)(\tau+u)^3\right]\left(e^{3v}-1\right)} \\ &- \frac{e^{6v}}{(\tau-3u)^3}\left[(\tau-u)^3e^{3v} - (\tau-3u)(\tau^2+3u^2)\right]\,.\end{aligned} \tag{17}$$

By substituting $t = \ln s$ with this solution of s, one obtains the desired relations $x = x(u,y)$ and $y = y(u,v)$ for the points over the x-axis in the xy-plane. Besides, from the existence conditions, it is found that u must satisfy the inequation $u \geq |\tau|/3$. One also obtains that v can take any real value. For this values of u and v, the square root of equation 16 takes real positive values, see Appendix X for a rigorous proof.

V. TRICONFOCAL OVALS

In the section II, it was mentioned that any oval is related to three special points called foci. Here we are going to use this fact to construct two families of curves that intersect each other at right angles. To see this, consider the equation that relates ξ to the other parameters of the oval

$$\xi = \frac{u^2 - \tau^2}{(1-a^2)\tau}\,. \tag{18}$$

From this it is possible to solve for u, so that equation 1 can be written as

$$r + ar' = \pm\sqrt{\xi\tau(1-a^2)+\tau^2}\,. \tag{19}$$

Squaring both sides and solving for a gives $a_\pm = \frac{-rr'\pm\tau r''}{r'^2+\xi\tau}$. This allows to define the variables

$$\alpha = \frac{-\sqrt{x^2+y^2}\sqrt{(x-\tau)^2+y^2}+\tau\sqrt{(x-\tau-\xi)^2+y^2}}{(x-\tau)^2+y^2+\xi\tau}\,, \tag{20}$$

$$\beta = -\frac{\sqrt{x^2+y^2}\sqrt{(x-\tau)^2+y^2}+\tau\sqrt{(x-\tau-\xi)^2+y^2}}{(x-\tau)^2+y^2+\xi\tau}\,. \tag{21}$$

The α- and β-constant curves are the set of ovals that share the same three foci, and hence they are called as *triconfocal* ovals. It is straight forward to show that this curves mutually orthogonally intersect each other, i.e., that $\nabla\alpha\cdot\nabla\beta = 0$. The property that triconfocal ovals intersect at right angles is also reported by Williamson in [14].

To find the inverse relations, $x = x(\alpha,\beta)$ and $y = y(\alpha,\beta)$, it is convenient to write

$$\alpha+\beta = -\frac{2rr'}{\xi\tau+r'^2}\,, \tag{22}$$

$$\alpha\beta = \frac{-\xi\tau-\tau^2+r^2}{\xi\tau+r'^2}\,. \tag{23}$$

From equation 22 it is found that $r = -(\alpha+\beta)(\xi\tau+r'^2)/2r'$. Replacing this in equation 23 and reorganizing one gets

$$(\beta-\alpha)^2 r'^4 + 2\left(-2\tau+\xi\left(\alpha^2+\beta^2-2\right)\right)\tau r'^2 + \xi^2\tau^2(\alpha+\beta)^2 = 0\,. \tag{24}$$

The two solutions of this equation, viewed as a quadric in $\mu \equiv r'^2$, are

$$\mu_\pm = \frac{\pm 2\sqrt{\tau^2\left(\beta^2\xi-\tau-\xi\right)\left(\alpha^2\xi-\tau-\xi\right)}+2\tau^2+\left(-\alpha^2-\beta^2+2\right)\xi\tau}{(\beta-\alpha)^2}\,. \tag{25}$$

Since $\mu_+ - \mu_- = 4\sqrt{\tau^2\left(\beta^2\xi-\tau-\xi\right)\left(\alpha^2\xi-\tau-\xi\right)}/(\beta-\alpha)^2 > 0$, one has $\mu_+ > \mu_-$. If the intersections of α- and β-constant curves occur at the same value of r', then μ_+ must be the correct solution to ensure $r'^2 > 0$. Taking $r'^2 = \mu_\pm$ it is obtained that

$$\begin{aligned} x = \Big[&\left(\pm\alpha\beta-1\right)\sqrt{\tau^2\left(\beta^2\xi-\tau-\xi\right)\left(\alpha^2\xi-\tau-\xi\right)} \\ &-\left[\left(1-\alpha^2+\alpha\beta-\beta^2\right)\tau+\xi\left(\alpha^2-1\right)\left(\beta^2-1\right)\right]\tau\Big]\frac{1}{(\beta-\alpha)^2\tau}\,, \end{aligned} \tag{26}$$

$$y^2 = -\left(\xi\tau(\alpha\beta-1)-\tau^2\mp\sqrt{\tau^2\left(\beta^2\xi-\tau-\xi\right)\left(\alpha^2\xi-\tau-\xi\right)}\right)^2\frac{\left(\alpha^2-1\right)\left(\beta^2-1\right)}{(\beta-\alpha)^4\tau^2}\,. \tag{27}$$

where it was considered the equations $x = \frac{r'^2-\tau^2-r^2}{2\tau}$ and $y^2 = r^2 - \frac{(r'^2-\tau^2-r^2)^2}{(2\tau)^2}$, derived from the definitions of r and r' in terms of x and y.

The coordinates just defined are indeed a family of coordinates that depend on τ and ξ. Some interesting cases are $\xi = 0$ and $\xi = -\tau$, with τ being any nonzero real number. For this cases, the corresponding ovals became limaçons of Pascal. This can be seen as follows. For $\xi = 0$, one has $\alpha = (-r + \tau)/r'$. By the law of cosines, $r = \sqrt{r'^2 + \tau^2 + 2r'\tau\cos\phi}$, see Fig. 1. Hence, one can find

$$r' = \frac{2\tau}{\alpha^2 - 1}(\cos\phi + \alpha)\,, \tag{28}$$

which is the equation of a limaçon in its standard form. Anagously for β, one finds $r' = \frac{2\tau}{\beta^2-1}(\cos\phi - \beta)$. Therefore, α- and β-constant curves are limaçons. For $\xi = -\tau$, one can proceed as before to find $r_\pm = \frac{2a\tau}{1-a^2}(\pm 1 - a\cos\theta)$, where the case r_+ is for $a = \alpha$ and the case r_- is for $a = \beta$. The fact that limaçons are particular cases of ovals is also mentioned in [12].

Hence, triconfocal oval coordinates can be reduced to *Pascal limaçon coordinates*. As far as the authors of this article knows, coordinates of this kind, which have limaçons as constant curves, have not been constructed before. For $\xi = 0$, equations 26 and 27 become $x = \tau$ and $y^2 = 0$ for the second sign. Since this is inconsistent, one must take first sign instead. By doing so, one gets

$$x = \frac{\tau(\alpha^2 + \beta^2 - 2)}{(\beta - \alpha)^2}\,, \tag{29}$$

$$y^2 = -\frac{4\tau^2(\alpha^2 - 1)(\beta^2 - 1)}{(\beta - \alpha)^4}\,. \tag{30}$$

This case may be considered as the simplest case of triconfocal oval coordinates.

Finally, viewing α and β as functions of x, y, ξ, and τ one finds that

$$\beta(x, y; \xi, \tau) = \alpha(-x, -y; -\xi, -\tau)\,, \tag{31}$$

$$\alpha(\tau x, \tau y; \tau\xi, \tau) = \begin{cases} \alpha(x, y; \xi, 1)\,, & \text{if } \tau > 0; \\ \beta(x, y; \xi, 1)\,, & \text{if } \tau < 0\,. \end{cases} \tag{32}$$

By analysing $\alpha(x, y; \xi, 1)$ and $\beta(x, y; \xi, 1)$, viewed as functions of x and y, one can find the range of values that α and β can take in the inverse relations (equations 26 and 27). Let $\mathcal{D}_\alpha$ and $\mathcal{D}_\beta$ be such set of values so that $\alpha \in \mathcal{D}_\alpha$ and $\beta \in \mathcal{D}_\beta$. Then, it is found that $\mathcal{D}_\alpha$ is $[-1, 1]$, if $\tau > 0$ and $\frac{\xi}{\tau} > -1$; $[-1, f\left(\frac{\xi}{\tau}\right)]$, if $\tau > 0$ and $\frac{\xi}{\tau} \leq -1$; $(-\infty, -1] \cup [1, \infty)$, if $\tau < 0$

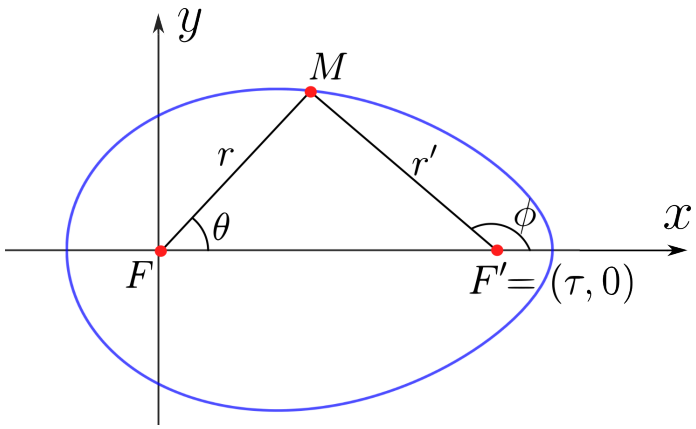


FIG. 1: A Descartes oval shown over an xy-plane choosen so that the foci lay on the x-axis. The equation of the curve is $r + ar' = u$, with $\tau = 1$, $a = 0.7$ and $u = 1.2$.

and $\frac{\xi}{\tau} < 0$; $(-\infty, -1]$ if $\tau < 0$ and $\frac{\xi}{\tau} = 0$; and $[-f\left(\frac{\xi}{\tau}\right), -1]$, if $\tau > 0$ and $\frac{\xi}{\tau} > 0$, where

$$f(\xi) = \frac{(\xi+1)\left(\sqrt{\xi(1+\xi)} - \xi\right)}{\xi\left(\xi + 1 - \sqrt{\xi(1+\xi)}\right)}, \tag{33}$$

is the value of $\alpha(x, y; \xi, 1)$ at $x = x_0$ and $y = 0$, with x_0 being so that $\frac{\partial\alpha(x,0)}{\partial x}\Big|_{x=x_0} = 0$ and $x_0 > 0$. Expressing the specific set $\mathcal{D}_\alpha$ for certain ξ and τ as $\mathcal{D}_\alpha(\xi, \tau)$, one has $\mathcal{D}_\beta(\xi, \tau) = \mathcal{D}_\alpha(-\xi, -\tau)$.

VI. CHARACTERIZATION OF THE VARIOUS ORTHOGONAL COORDINATES

Confocal and triconfocal ovals induced certain systems of coordinates in which the coordinate curves are one of these class of ovals. In this section we will characterize these coordinates in terms of their metric coefficients and their basis vectors. From the methods mentioned in [5], 3D coordinates will be constructed from the 2D coordinates by "rotating" or "translating" the coordinates. This process results in cylindrical- and rotational-like coordinates, respectively.

Consider confocal ovals as u-constant curves and synecdolas as v-constant ones. The quantities u and v define a set of *confocal oval 2D coordinates*, or simply *oval coordinates*. The metric coefficients are described in terms of the functions $x(u, v) \equiv \Lambda(u, v)$ and $y(u, v) \equiv$

$\Upsilon(u,v)$ as

$$ds^2 = \left(\Lambda_u^2 + \Upsilon_u^2\right)\left(du^2 + \Upsilon^2 dv^2\right), \tag{34}$$

where the subscripts indicates partial differentiation. These function satisfy the relations

$$\Lambda_u = -\frac{\sqrt{\Lambda^2+\Upsilon^2}\,a\,(\tau-\Lambda) - \Lambda\sqrt{(\Lambda-\tau)^2+\Upsilon^2}}{2a\left(\Lambda^2+\Upsilon^2-\Lambda\tau\right) + \left(a^2+1\right)\sqrt{\Lambda^2+\Upsilon^2}\sqrt{(\Lambda-\tau)^2+\Upsilon^2}}, \tag{35}$$

$$\Upsilon_u = \frac{\Upsilon\left(\sqrt{\Lambda^2+\Upsilon^2}\,a + \sqrt{(\Lambda-\tau)^2+\Upsilon^2}\right)}{2a\left(\Lambda^2+\Upsilon^2-\Lambda\tau\right) + \left(a^2+1\right)\sqrt{\Lambda^2+\Upsilon^2}\sqrt{(\Lambda-\tau)^2+\Upsilon^2}}, \tag{36}$$

$$\Lambda_v = \Upsilon\Upsilon_u\,, \tag{37}$$

$$\Upsilon_v = -\Upsilon\Lambda_u\,, \tag{38}$$

which can be obtained by taking partial differentiation to the equations $u = \sqrt{x^2+y^2} + a\sqrt{(x-\tau)^2+y^2}$ and $v = \operatorname{arcsinh}(x/y) + a\operatorname{arcsinh}((x-\tau)/y)$, which define u and v in terms of x and y. In addition, the basis vectors related to u and v are

$$\hat{\boldsymbol{e}}_u = \frac{1}{\sqrt{\Lambda_u^2+\Upsilon_u^2}}\left(\Lambda_u\hat{\boldsymbol{e}}_x + \Upsilon_u\hat{\boldsymbol{e}}_y\right), \tag{39}$$

$$\hat{\boldsymbol{e}}_v = \frac{1}{\sqrt{\Lambda_u^2+\Upsilon_u^2}}\left(\Upsilon_u\hat{\boldsymbol{e}}_x - \Lambda_u\hat{\boldsymbol{e}}_y\right). \tag{40}$$

Another system of coordinates may be derived from triconfocal ovals. The quantities α and β, defined in the previous section, are consider as a set of *triconfocal oval 2D coordinates*, or *triconfocal coordinates* for short. In this case, the metric coefficients are given by

$$ds^2 = g_{\alpha\alpha}\left(d\alpha^2 + \frac{(\alpha^2-1)(\alpha^2\xi-\tau-\xi)}{(1-\beta^2)(\beta^2\xi-\tau-\xi)}d\beta^2\right), \tag{41}$$

with

$$g_{\alpha\alpha} = \Big[2\left(-\alpha\beta\xi+\tau+\xi\right)\sqrt{\tau^2\left(\beta^2\xi-\tau-\xi\right)\left(\alpha^2\xi-\tau-\xi\right)} + \big(2\left(\alpha^2\beta^2+1\right)\xi - \left(\beta+\alpha\right)^2\left(\xi+\tau\right) - 4\tau\big)\xi\tau + 2\tau^3\Big]\frac{\beta+\alpha}{\left(\alpha^2-1\right)\left(\beta-\alpha\right)^3\left(\alpha^2\xi-\tau-\xi\right)}. \tag{42}$$

The basis vectors are ... In the case $\xi = 0$, this coordinates get reduced to Pascal limaçon coordinates, whose metric coefficients are given by

$$ds^2 = \frac{4(\alpha+\beta)\tau^2}{(\alpha-\beta)^3}\left(\frac{1}{\alpha^2-1}d\alpha^2 + \frac{1}{1-\beta^2}d\beta^2\right). \tag{43}$$

Based on the two set of coordinates described before, one can easily obtain cylindrical-like coordinates. For this one just needs to introduce a z coordinate in the orthogonal direction

to the xy-plane in which the 2D coordinate curves live. Hence, one obtains *confocal oval-cylinder coordinates* u, v and z, and *triconfocal oval-cylinder coordinates* α, β and z. Their line elements are respectively

$$ds^2 = \left(\Lambda_u^2 + \Upsilon_u^2\right)\left(du^2 + \Upsilon^2 dv^2\right) + dz^2\,, \tag{44}$$

$$ds^2 = g_{\alpha\alpha}\left(d\alpha^2 + \frac{(\alpha^2-1)(\alpha^2\xi-\tau-\xi)}{(1-\beta^2)(\beta^2\xi-\tau-\xi)}d\beta^2\right) + dz^2\,. \tag{45}$$

It should be pointed out that the Laplace equation admits a separable solution for the triconfocal oval-cylinder coordinates, but not for the confocal ones.

Next, rotational coordinates can be obtained in the next way. Consider a point $(x(q^1,q^2), y(q^1,q^2), 0)$ in some xyz-space related to coordinates q^1 and q^2. Rotate this point around either the x- or y-axis in a certain angle ϕ. The set of points obtained by fixing one of the quantities q^1, q^2 and ϕ define a certain surface. The resulting classes of surfaces orthogonally intersect each other and define the coordinate surfaces of certain system of coordinates. Naturally, it only make sense to rotate the points around the symmetry axis of the coordinate 2D curves. For ovals, this coincides with the focal axis.

Applying this to confocal ovals, one obtains the coordinates

$$\tilde{u} = \sqrt{x^2+y^2+z^2} + a\sqrt{x^2+y^2+(z-\tau)^2}\,, \tag{46}$$

$$\tilde{v} = \operatorname{arcsinh}\left(\frac{z}{\sqrt{x^2+y^2}}\right) + a\operatorname{arcsinh}\left(\frac{z-\tau}{\sqrt{x^2+y^2}}\right)\,, \tag{47}$$

$$\phi = \operatorname{atan2}(y,x)\,, \tag{48}$$

with inverse relations $x = \Upsilon(\tilde{u},\tilde{v})\cos\phi$, $y = \Upsilon(\tilde{u},\tilde{v})\sin\phi$ and $z = \Lambda(\tilde{u},\tilde{v})$. The corresponding line element is

$$ds^2 = \left(\Lambda_{\tilde{u}}^2 + \Upsilon_{\tilde{u}}^2\right)\left(d\tilde{u}^2 + \Upsilon^2 d\tilde{v}^2\right) + \Upsilon^2 d\phi^2\,. \tag{49}$$

One can check that $\tilde{u}-$ and $\tilde{v}-$constant surfaces are surfaces of revolution of their respective coordinate curves around the symmetry axis. Since the surface of revolution of ovals are known as ovoids, this coordinates may be called as *confocal ovoid 3D coordinates*, or simply as *ovoid coordinates*. In the same spirit, the term *synecdoloids* is coined to refer to the $\tilde{v}$-constant surfaces, in analogy to the one used for the surface of revolution of hyperbolas. The line element 49 is the one that will be use in the next section to construct stationary spacetimes with ovoid symmetry.

Finally, for triconfocal oval coordinates one obtains

$$\tilde{\alpha} = \frac{-\sqrt{z^2+x^2+y^2}\sqrt{(z-\tau)^2+x^2+y^2}+\tau\sqrt{(z-\tau-\xi)^2+x^2+y^2}}{(z-\tau)^2+x^2+y^2+\xi\tau}\,, \tag{50}$$

$$\tilde{\beta} = -\frac{\sqrt{z^2+x^2+y^2}\sqrt{(z-\tau)^2+x^2+y^2}+\tau\sqrt{(z-\tau-\xi)^2+x^2+y^2}}{(z-\tau)^2+x^2+y^2+\xi\tau}\,. \tag{51}$$

$$\phi = \mathrm{atan2}(y,x)\,, \tag{52}$$

with the inverse relations $x = \kappa(\tilde{\alpha},\tilde{\beta})\cos\phi$, $y = \kappa(\tilde{\alpha},\tilde{\beta})\sin\phi$ and $z = \eta(\tilde{\alpha},\tilde{\beta})$, where $\eta(\alpha,\beta)$ is taken to be the right hand side of the equation 26 and $\kappa(\alpha,\beta)$ is taken to be the square root of the right hand side of the equation 27. The quantity ϕ takes values in the interval $[0,2\pi)$, and as in the previous section $\tilde{\alpha}\in\mathcal{D}_{\tilde{\alpha}}$ and $\tilde{\beta}\in\mathcal{D}_{\tilde{\beta}}$. The line element is

$$ds^2 = g_{\tilde{\alpha}\tilde{\alpha}}\left(d\tilde{\alpha}^2 + \frac{(\tilde{\alpha}^2-1)(\tilde{\alpha}^2\xi-\tau-\xi)}{(1-\tilde{\beta}^2)(\tilde{\beta}^2\xi-\tau-\xi)}d\tilde{\beta}^2\right)+\kappa^2 d\phi^2\,. \tag{53}$$

This coordinates may be called as *triconfocal ovoid 3D coordinates*.

TABLE I: Orthogonal coordinates. $\mathcal{D}_u$ is given by the equation 73 and $\mathcal{D}_\alpha$ and $\mathcal{D}_\beta$ are described in the section V. For the Pascal limaçon coordinates it was made $\alpha\equiv\mu$ and $\beta\equiv\nu$.

System of coordinates	Coordinates	Conditions		
Confocal oval coordinates	u,v	$u\in\mathcal{D}_u$	$v\in\mathbb{R}$	
Confocal oval-cylinder coordinates	u,v,z	$u\in\mathcal{D}_u$	$v\in\mathbb{R}$	$z\in\mathbb{R}$
Confocal ovoid coordinates	$\tilde{u},\tilde{v},\phi$	$\tilde{u}\in\mathcal{D}_u$	$\tilde{v}\in\mathbb{R}$	$\phi\in[0,2\pi)$
Triconfocal oval coordinates	α,β	$\alpha\in\mathcal{D}_\alpha$	$\beta\in\mathcal{D}_\beta$	
Pascal limaçon coordinates	μ,ν	$\mu\in[-1,1]$	$\nu\in(-\infty,-1]$	
Triconfocal oval-cylinder coordinates	α,β,z	$\alpha\in\mathcal{D}_\alpha$	$\beta\in\mathcal{D}_\beta$	$z\in\mathbb{R}$
Triconfocal ovoid coordinates	$\tilde{\alpha},\tilde{\beta},\phi$	$\tilde{\alpha}\in\mathcal{D}_\alpha$	$\tilde{\beta}\in\mathcal{D}_\beta$	$\phi\in[0,2\pi)$

VII. OVOID SPACETIMES

The symmetries in a spacetime are usually characterized by geometric invariants such as Killing vectors. These vector can be associated to a group of isometries isomorphic to a certain group such as $SO(2)$ or $SO(3)$. In this sense, it is natural to define an axially

symmetric spacetime or a spherically symmetric spacetime as one in which there is a subgroup G of the group of isometries[1] which is isomorphic to $SO(2)$ or $SO(3)$, respectively. However, for other spacetimes such as ellipsoidal or ovoid spacetimes, there is not such a natural definition since there is not a group related to ellipsoids or ovoids. The best one can do is to propose a metric ansatz that encapsulates some desirable features. One of such may be that, for example, there exits some congruence of observers in which the induced metric over their orthogonal hypersurfaces resembles or be similar to the metric over ellipsoids or ovoids. Such proposal is the one of Krasiński [8], which construct an stationary oblate-ellipsoidal metric ansatz. Here, a stationary ovoid metric ansatz will be built based on the same procedure.

In order to do that, consider a congruence of observers characterized by the four-velocity $w = U\partial_t + V\partial_\phi$. As the spacetime is consider to be stationary and axially symmetric, there exist two conmuting killing vector fields, which here are considered as ∂_t and ∂_ϕ. The projection of the metric tensor over the orthogonal hypersurfaces of w is taken as

$$(g_{ab}+w_a w_b)dx^a\otimes dx^b = f^2(\Lambda_u^2+\Upsilon_u^2)du\otimes du+k^2\left[\Upsilon^2(\Lambda_u^2+\Upsilon_u^2)dv\otimes dv+\Upsilon^2 d\Phi\otimes d\Phi\right]\,, \quad (54)$$

where f and k are functions of u and v, and $d\Phi = d\phi - (V/U)dt$. From here one can deduce the most general form of a metric tensor that satisfies the equation 54, which is

$$\begin{aligned}\mathcal{G} = g_{00}dt\otimes dt + g_{03}(dt\otimes d\phi + d\phi\otimes dt) + A^2 d\phi\otimes d\phi + f^2(\Lambda_u^2+\Upsilon_u^2)du\otimes du \\ + k^2\Upsilon^2(\Lambda_u^2+\Upsilon_u^2)dv\otimes dv\,. \end{aligned} \quad (55)$$

Over the 2-surfaces spanned by the vectors ∂_v and ∂_ϕ, the induced metric is conformal to that of an ovoid (see equation 49). To simplify the notation, the tilde emphasis in u and v has been omitted. Equation 54 shows that there is a part of the induced metric that is conformal to the metric over confocal ovoids, with k^2 being the conformal factor. The same procedure describe before can be implemented to derive a triconfocal ovoid metric ansatz $\mathcal{G}'$. For simplicity, this article focus only on this confocal case. Now it is convenient to write the static case of the metric $\mathcal{G}$ because it will be useful later. This corresponds to the metric

$$\mathcal{G}_{static} = g_{00}dt\otimes dt + k^2\Upsilon^2 d\phi\otimes d\phi + f^2(\Lambda_u^2+\Upsilon_u^2)du\otimes du + k^2\Upsilon^2(\Lambda_u^2+\Upsilon_u^2)dv\otimes dv\,. \quad (56)$$

[1] $\mathrm{Iso}(M,\mathcal{G}) := \{\phi : M\to M \mid \phi^*\mathcal{G} = \mathcal{G}\}$ endowed with the composition operation, where M is the manifold, $\mathcal{G}$ is the metric tensor and ϕ^* is the pushforward of ϕ.

It should be mention that by considering the equations 35 and 36 to write Λ_u and Υ_u in terms of $\Lambda = z$ and $\Upsilon = \sqrt{x^2+y^2}$ and by fixing $a = 0$ or $\tau = 0$, $\mathcal{G}_{static}$ in spherical coordinates is reduced to

$$\mathcal{G}_{spheric} = g_{00}dt \otimes dt + k^2r^2\sin^2\theta\, d\phi \otimes d\phi + f^2 dr \otimes dr + k^2r^2 d\theta \otimes d\theta\,. \tag{57}$$

which is a suitable metric ansatz to describe static spherically symmetric spacetimes. Hence, the *ovoid spacetimes* here described have the spherically symmetric spacetimes as a particular case. The condition $a = 0$ or $\tau = 0$ makes the u-constant surfaces in the ovoid coordinate system become concentric spheres to the origin with radius $r = u$ or $r = u/(1+a)$, respectively.

VIII. ANYSOTROPIC FLUID SOLUTIONS TO THE EINSTEIN'S FIELD EQUATIONS

The relevance to define spacetimes with new symmetries is that it opens the possibility to obtain new exact solutions to the Einstein field equations. If it is a vacuum solution, it can be interpreted as the solution for a black hole ore one part of it can be interpreted as the external solution for an object with ovoid symmetry. And, if it is a fluid solution, one would obtain a model for the gravitational field inside an object with ovoid symmetry. Both cases are very interesting and stimulating. Sadly, the Einstein field equations get too complicated. For this article, only a static anysotropic fluid solution is obtained.

Consider a static ovoid spacetime govern by the metric $\mathcal{G}_{static}$. A suitable stress-energy tensor for an static anysotropic fluid is

$$\mathcal{T} = T^{ab}\partial_a \otimes \partial_b = (\mu + P_1)w \otimes w + P_1\mathcal{G}^{-1}_{static} + (P_2 - P_1)e_2 \otimes e_2 + (P_3 - P_1)e_3 \otimes e_3\,, \tag{58}$$

where w is the four-velocity of the fluid, μ is its mass-energy density, P_i is its pressure in the direction of e_i with $i = 1, 2, 3$, $\{e_\mu\} = \{w, e_1, e_2, e_3\}$ is the orthonormal tetrad of the observer with four-velocity w, and $\mathcal{G}^{-1}_{static}$ is the inverse metric. Naturally, the quantities μ and P_i are the ones measured by the comoving observer with the fluid. In the case $P_2 = P_3 = P_1$ one recovers the stress-energy tensor of a perfect fluid in its standard form. In natural units the Einstein field equations are

$$R_{ab} = 8\pi\left(T_{ab} - \frac{1}{2}Tg_{ab}\right)\,, \tag{59}$$

where T is the trace of $\mathcal{T}$ and $\mathcal{R} = R_{ab}\, dx^a \otimes dx^b$ is the Ricci tensor.

A solution to this equations is given by

$$P_1 = \frac{1}{8\pi\Upsilon^2(\Lambda_u^2+\Upsilon_u^2)^2}\left(\mathcal{P} - A_u^2\Upsilon^2(\Lambda_u^2+\Upsilon_u^2)e^{-2B}\right), \tag{60}$$

$$P_2 = \frac{1}{8\pi\Upsilon^2(\Lambda_u^2+\Upsilon_u^2)^2}\left\{\mathcal{L} - \left[(A_u+B_u)\Upsilon_u - A_u^2\Upsilon\right](\Lambda_u^2+\Upsilon_u^2)\Upsilon e^{-2B}\right\}, \tag{61}$$

$$P_3 = \frac{1}{8\pi\Upsilon^2(\Lambda_u^2+\Upsilon_u^2)^3}\left\{\mathcal{Q} + \left[A_u^2\Upsilon^2(\Lambda_u^2+\Upsilon_u^2) - \mathcal{L}\right](\Lambda_u^2+\Upsilon_u^2)e^{-2B}\right\}, \tag{62}$$

$$\begin{aligned}\mu = \frac{1}{8\pi\Upsilon^2(\Lambda_u^2+\Upsilon_u^2)^3}\Big\{ &- \mathcal{Q} + \big[2\Upsilon^2 A_{uu} + (B_u+5A_u)\Upsilon\Upsilon_u \\ &- A_u\Upsilon^2(2B_u+3A_u) + \mathcal{P}\big](\Lambda_u^2+\Upsilon_u^2)^2 e^{-2B}\Big\},\end{aligned} \tag{63}$$

$$g_{00} = e^{2A}, \tag{64}$$

$$f = e^{2B}, \tag{65}$$

$$k = e^{2(c_1-A)}, \tag{66}$$

where A and B are arbitrary functions of u, c_1 is a constant, and $\mathcal{P}$, $\mathcal{Q}$ and $\mathcal{L}$ are given by

$$\mathcal{P} = \Upsilon_u\left(\Upsilon_{uv}\Lambda_u - \Lambda_{uv}\Upsilon_u\right)\left(e^{2(A-c_1)} - e^{-2B}\right), \tag{67}$$

$$\begin{aligned}\mathcal{Q} = &\left[(\Lambda_u^2-\Upsilon_u^2)(\Upsilon_{uv}^2+\Lambda_{uv}^2) + (\Lambda_u\Lambda_{uvv} + \Upsilon_u\Upsilon_{uvv})(\Lambda_u^2+\Upsilon_u^2) - 4\Lambda_u\Upsilon_u\Lambda_{uv}\Upsilon_{uv}\right] \\ &\times\left(e^{2(A-c_1)} - e^{-2B}\right) + (A_u+B_u)(\Lambda_u^2+\Upsilon_u^2)(\Lambda_u\Upsilon_{uv} - \Upsilon_u\Lambda_{uv})\Upsilon e^{-2B},\end{aligned} \tag{68}$$

$$\mathcal{L} = \Lambda_u\left(\Lambda_{uv}\Lambda_u + \Upsilon_{uv}\Upsilon_u\right)\left(e^{-2B} - e^{2(A-c_1)}\right), \tag{69}$$

In order to check that this solutions satisfies the equation 59 it is convenient to use the relations shown in the section X D to simplify the expressions that involve derivatives of Λ and Υ. In this scenario, the weak and strong energy conditions can be satisfied if $\mu, P_i \geq 0$. In the limit when $x, y, z \gg \tau$ this inequalities take the approximate form

$$uA_u^2 - (A_u + B_u) \geq 0, \tag{70}$$

$$1 - e^{2(A+B-c_1)} - u^2A_u^2 \geq 0, \tag{71}$$

$$2A_u - uA_uB_u + uA_{uu} - uA_u^2 \geq 0. \tag{72}$$

These were obtained by expressing the derivatives of Λ and Υ in terms of themselves (see the appendix X D) and taking the high order terms in x, y and z. A solution to the exact or the approximate energy conditions is left for a future work.

IX. CONCLUSIONS

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X. APPENDIX

A. Existence condition for Descartes ovals

The existence conditions of ovals may be simplified by introducing the set $\mathcal{D}_u$ given by

$$\mathcal{D}_u(a;\tau) := \begin{cases} [\min(1,a)|\tau|, \infty) & \text{if } a > -1\,; \\ [-|\tau|, |\tau|] & \text{if } a = -1\,; \\ (-\infty, |\tau|] & \text{if } a < -1\,. \end{cases} \tag{73}$$

B. Orthogonality condition for coordinates q_1, q_2

Here, it is shown that any two coordinates q_1, q_2 are orthogonal if and only if their coordinate curves meet at right angles. To do so, consider the relations $q^i = q^i(x,y)$ and their gradient $\boldsymbol{\nabla} q^i$. Since $\boldsymbol{\nabla} q^i$ is orthogonal to the level curves of q^i then the coordinate curves intersect at right angles if

$$\boldsymbol{\nabla} q^1 \cdot \boldsymbol{\nabla} q^2 = q^1_x q^2_x + q^1_y q^2_y = 0\,, \tag{74}$$

where the subscripts indicate partial differentiation, $q^i_x := \partial q^i/\partial x$ for each $i = 1, 2$. By the chain rule one has that

$$\frac{\partial q^i}{\partial q^j} = q^i_x x_{,j} + q^i_y y_{,j}\,, \tag{75}$$

for each $i = 1, 2$ and $j = 1, 2$. For $i \neq j$, $\partial q^i/\partial q^j = 0$. Hence, in this case, one gets that $q^i_x = -q^i_y y_{,j}/x_{,j}$, which implies that $\boldsymbol{\nabla} q^1 \cdot \boldsymbol{\nabla} q^2 = q^1_y q^2_y (y_{,1}y_{,2} + x_{,1}x_{,2})/x_{,1}x_{,2}$, so that equation 8 implies equation 74, and vice versa.

C. The s-equation as an algebraic equation in some power of s

D. Characterization of the second and third derivatives of the metric functions Λ and Υ